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Retraining, Fine-Tuning and Grad-CAM Analysis of a VGG19-Based CNN

1. Introduction

Facial Expression Recognition (FER) has become an important research area within computer vision and deep learning due to its wide range of applications in human-computer interaction, healthcare, education, security, and affective computing. However, emotion recognition is a hard task, even for humans. What for one person is a surprised face, for another it may be a scared one.

The goal of this project is to evaluate how good a Convolutional Neural Network (CNN) for facial expressions recognition is. To achieve this we will be using a VGG19-based CNN created by WuJie in the year 2018. This model was created for a challenge, and was created to train and test on CK+ and FER13 datasets, where it achieved significant results.

In our project we will focus on the following tasks. We will retrain the model with the same datasets as WuJie, do a Grad-CAM analysis to check the important features, and then test and fine-tune on a distinct dataset.

2. State of the art

Several CNN architectures have been applied to FER tasks. Where deeper architectures like VGG or ResNet achieved the best performances. Among these, VGG19 remains a popular choice due to its simple architecture, strong feature extraction capabilities, and suitability for transfer learning applications.

One of the most widely used FER datasets is CK+ (Extended Cohn-Kanade Dataset), which contains posed facial expressions captured under controlled laboratory conditions. Models trained on CK+ often achieve very high accuracy, frequently exceeding 95%, due to the limited variability and clear emotional expressions present in the dataset. However, these results often overestimate real-world performance because the dataset does not reflect natural human behavior.

To address this limitation, researchers increasingly evaluate models on more challenging datasets such as FER2013. This dataset contains 35,887 grayscale facial images collected from the internet and includes significant variations in lighting, pose, image quality, expression intensity and noise amount. For this reason FER2013 has become one of the standard benchmarks for facial expression recognition in unconstrained environments. State-of-the-art CNN-based approaches typically achieve accuracies between 70% and 75% on this dataset, which is close.

Compared to them, WuJie's CNN, that achieves an accuracy of 94,46% in CK+ and 73,112% in FER2013, well in the range of the state of the art of 2018.

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3. Methodology

The experimental work in this project follows a single sequential pipeline in which each stage is motivated by the outcome of the previous one. First, the VGG19-based CNN is retrained following WuJie's protocol on CK+ and FER2013 in order to reproduce a reliable baseline and confirm that the model behaves as reported in the original work. Once a baseline checkpoint is obtained, Grad-CAM is applied to interpret which facial regions drive the predictions. Because the Grad-CAM maps revealed an over-reliance on specific regions, an occlusion experiment is then run to quantify how much the model actually depends on those regions. The model is afterwards evaluated on RAF-CE in a zero-shot setting to measure how well the FER2013 representations transfer to a harder, real-world dataset containing compound emotions. Finally, given the limited zero-shot transfer, the model is fine-tuned on a combined FER2013 and RAF-CE set to improve cross-dataset generalization.
This ordering is deliberate. The interpretability (Grad-CAM) and robustness (occlusion) analyses are performed on the frozen baseline so that the conclusions describe the original model rather than an adapted one, and the zero-shot evaluation is carried out before any fine-tuning so that it provides a clean lower bound against which the benefit of fine-tuning can be measured. The datasets, architecture, and optimization details used at each of these stages are described in the following sections.

4. Data Analysis

For this project we used three distinct datasets, CK+, FER2013 and RAFCE.

CK+ was not touched, and was used as given in WuJie's repository. CK+ was split in two sets, one for training, and another for testing, just as for the original model.

FER2013 was the main dataset used for training and evaluation. It contains 35,887 grayscale facial images (48×48 pixels) labeled into seven emotion categories: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

Following WuJie's implementation, the original CSV file was converted to HDF5 format and images were reshaped and replicated into three RGB channels to match the VGG19 input requirements. During training, data augmentation was applied through a random crop to 44×44 pixels and a random horizontal flip (probability 0.5), with pixel values scaled to the [0,1] range; no mean/standard-deviation normalization was applied. TenCrop Test-Time Augmentation (TTA) was used during evaluation to improve prediction stability.

FER2013 provides a predefined split consisting of a training set (28,709 images), a PublicTest set used for model validation and hyperparameter selection (3,589 images), and a PrivateTest set used for final evaluation (3,589 images). The final model was selected according to its performance on the PrivateTest set.

RAF-CE was used to evaluate model generalization and for subsequent fine-tuning. Unlike CK+ and FER2013, it contains real-world images with compound emotions and greater variability in pose, lighting, occlusions, and subject demographics.

The FER2013-trained model was first evaluated in a zero-shot setting, mapping the 14 RAF-CE compound emotion classes to the closest FER2013 basic emotions. Afterwards, FER2013 and RAF-CE were combined for fine-tuning using a WeightedRandomSampler to address class imbalance.

Fine-tuning was performed progressively by first training the classifier, then unfreezing the last convolutional block (Block 5) together with the classifier head, and finally continuing to train that same block and head at a further reduced learning rate, always with decreasing learning rates.

5. Models and Optimization

For the model used we used the same VGG19 used by WuJie, which consists of 5 convolutional blocks and ending at a fully connected layer for classification.

- Block 1: 2 * Conv + BN + ReLU, MaxPool - 64 channels
- Block 2: 2 * Conv + BN + ReLU, MaxPool - 128 channels
- Block 3: 4 * Conv + BN + ReLU, MaxPool - 256 channels
- Block 4: 4 * Conv + BN + ReLU, MaxPool - 512 channels
- Block 5: 4 * Conv + BN + ReLU, MaxPool - 512 channels

Each convolution uses kernels of size 3 and padding of size 1. Additionally the model uses dropout with a probability of 0,5 and uses Cross entropy loss to classify the multiple categories available. For learning rate it uses a value of 0.01 with decay starting after epoch 80, where it is multiplied by a factor of 0.9 (a 10% reduction) every 5 epochs. The model is optimized with SGD (momentum 0.9, weight decay 5e-4) using a batch size of 64, and gradient clipping with a maximum norm of 0.1 is applied at every training step.

6. Experiments

First, the original VGG19 architecture from WuJie's repository was retrained on the FER2013 dataset following the same training protocol as the original implementation, which consisted of a 250-epoch training schedule. However, during the experimental process, only the metrics corresponding to the first 30 epochs were successfully preserved in the training log. For this reason, the training curves and epoch-level analysis reported in this work are limited to those first 30 epochs. The best model checkpoint obtained during training was then selected and used as the baseline model for the subsequent interpretability, robustness, and cross-dataset generalization experiments. Evaluation was performed using the same TenCrop testing strategy adopted in the original implementation.

Then, Grad-CAM (Gradient-weighted Class Activation Mapping) was applied to inspect which facial regions contributed most strongly to the model's predictions. Grad-CAM generates a heatmap by using the gradients flowing into the last convolutional layer, highlighting the image regions that most influence a specific classification decision.

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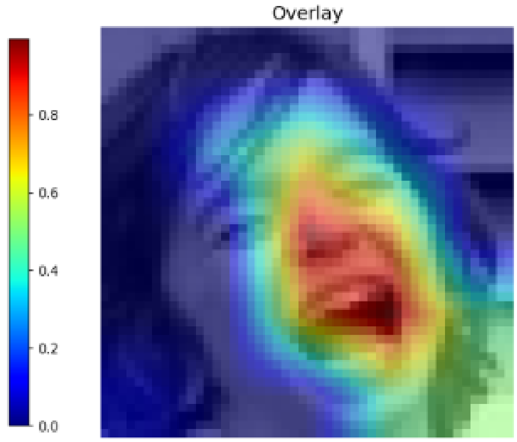

Figure 1: Grad-CAM example

The visualizations showed that the model generally focused on relevant facial areas such as the mouth, eyes, eyebrows, and central face structure. This experiment helped assess model interpretability and provided qualitative evidence that the network was not relying only on background artifacts, although some predictions still appeared sensitive to localized regions.

Since the Grad-CAM analysis highlighted the importance of specific facial regions for the model's predictions, an occlusion experiment was conducted to evaluate robustness. Artificial occlusions were introduced into the FER2013 test images by applying a centered black-square mask during evaluation, without modifying the original dataset.

Figure 2: Occlusion example

This approach allowed us to assess how the model's performance and decision-making were affected when key facial features were partially hidden, providing additional insight into the reliability of the learned visual representations.

Afterwards, the FER2013-trained model was evaluated on the RAF-CE dataset without any additional training in order to assess its cross-dataset generalization capabilities. Since RAF-CE contains compound emotion categories that do not directly match the seven basic emotions used in FER2013, its labels were mapped to the closest corresponding FER2013 classes. This zero-shot evaluation was designed to analyze the model's ability to transfer learned representations to a different dataset with more complex emotional expressions and greater real-world variability.

Finally, the model was fine-tuned using a mixed FER2013 and RAF-CE training strategy- to improve cross-dataset generalization while minimizing catastrophic forgetting. Fine-tuning was performed progressively in three stages: classifier-only training with the convolutional features frozen, partial fine-tuning that unfroze the final convolutional block (Block 5) together with the classifier head at a learning rate of 1e-4, and a final stage that continued training the same Block 5 and classifier head at a further reduced learning rate of 1e-5, with Blocks 1–4 kept frozen throughout. For fine-tuning the optimizer was also switched from SGD to Adam (weight decay 1e-4). A mixed training set was sampled using a WeightedRandomSampler to address class imbalance, and the best checkpoint was selected based on the combined validation performance across both datasets. This experiment was designed to adapt the model to compound emotions while preserving the facial representations previously learned from FER2013.

7. Results

Model trained with CK+ dataset

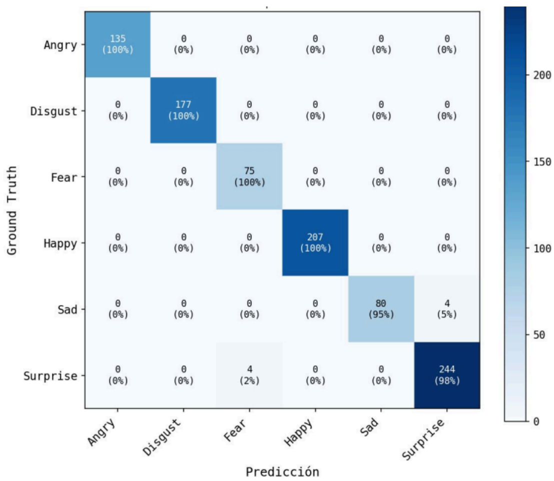

Table 1: Confusion Matrix

Model trained with FER2013 dataset:

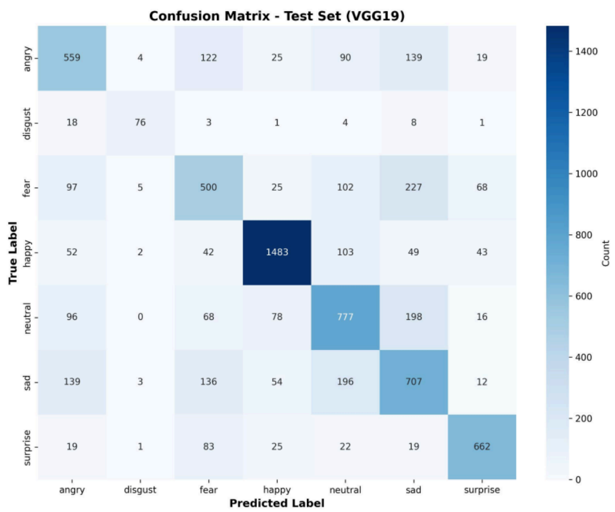

Table 2: Confusion Matrix

Class	Original
Happy	87.71%
Sad	85.58%
Disgust	78.18%
Surprise	74.76%
Angry	65.58%
Neutral	59.43%
Fear	56.06%

Table 3: Accuracy Table

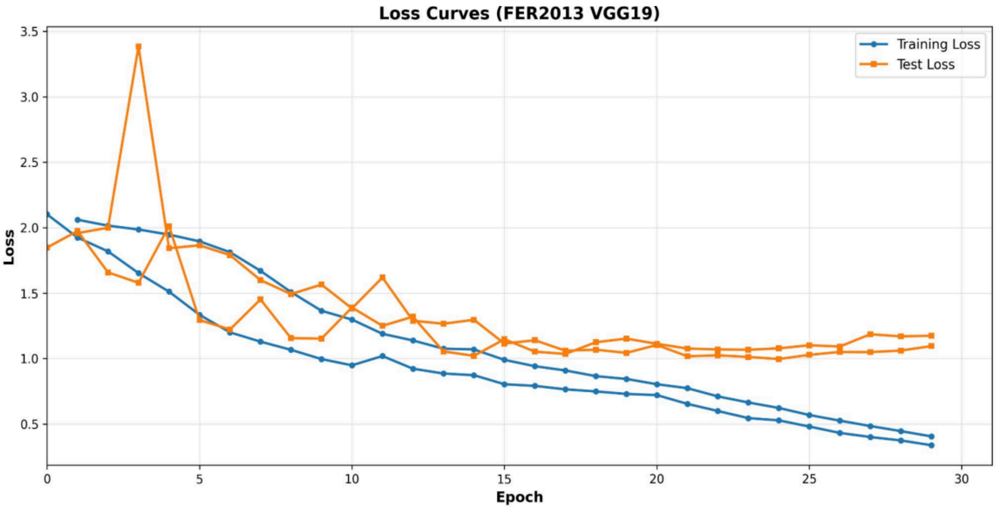

Figure 3: Loss Plot, Curves for Public and Private sets

Grad-CAM analysis:

Ground-Truth: Happy

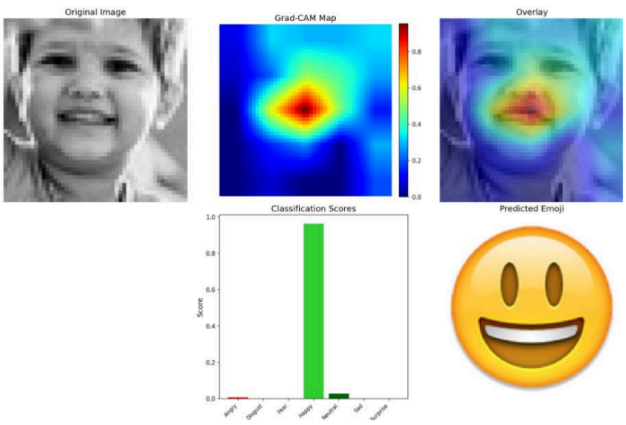

Figure 4: Correct classification

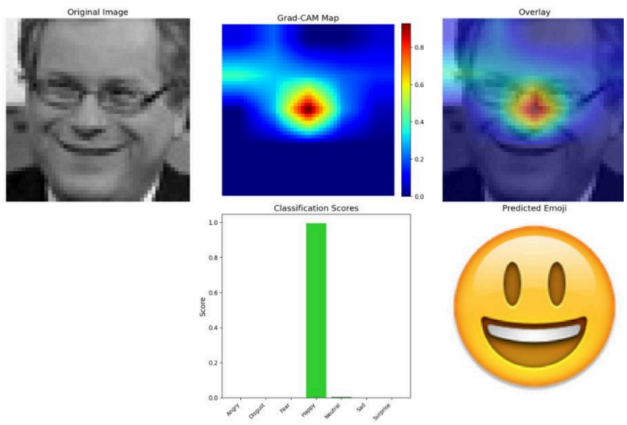

Figure 5: Correct classification

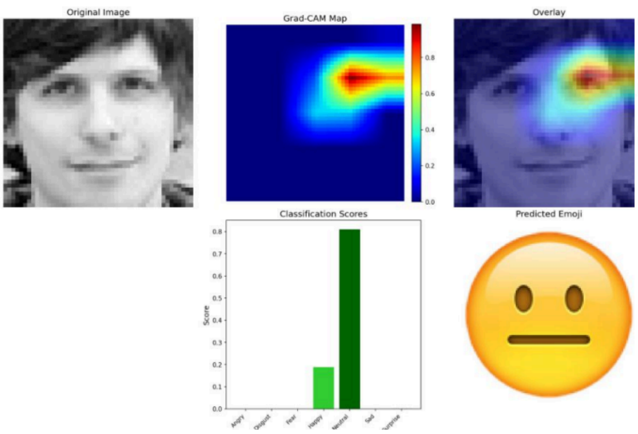

Figure 6: Incorrect classification

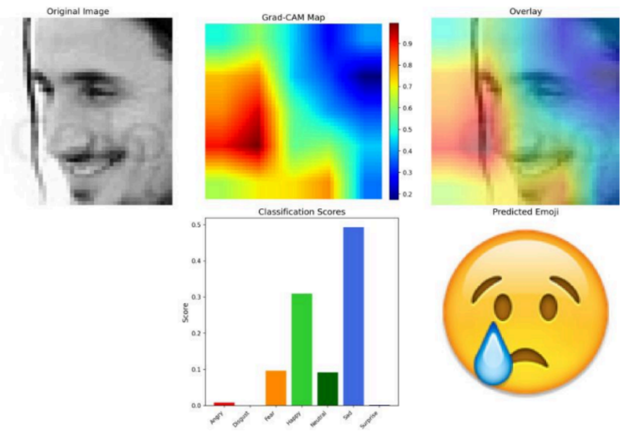

Figure 7: Incorrect classification

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Occlusion Experiment:

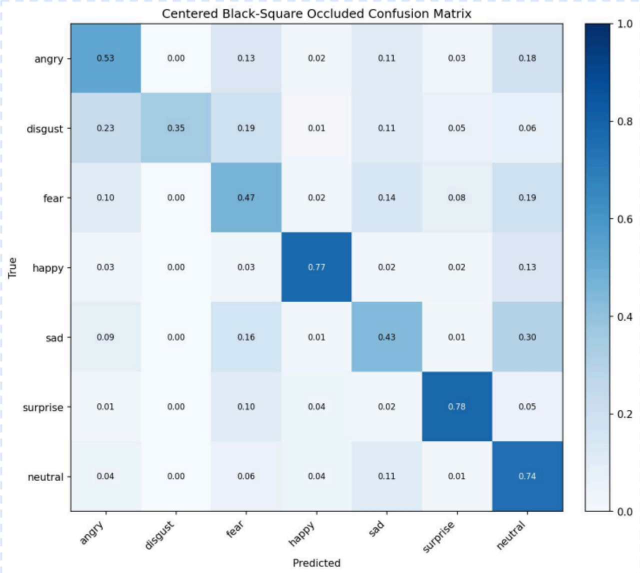


Table 4: Confusion Matrix

Class	Original	Occluded	Δ
Happy	87.71%	77%	-10.71%
Sad	85.58%	43.30%	-42,28%
Disgust	78.18%	35.14%	-43,04%
Surprise	74.76%	77.98%	+3,22%
Angry	65.58%	52.51%	-13,07%
Neutral	59.43%	74.37%	+17,94%
Fear	56.06%	46.58%	-9,48%

Table 5: Accuracy comparison

Fine-Tuning results:

Before Fine-Tuning

14.32% Loss: 6.792

Class	Acc.
Angry	39.22%
Disgust	7.36%
Fear	17.95%
Happy	35.00%
Sad	0.71%
Neutral	0%
Surprised	0%

After Fine-Tuning

30.31% Loss: 1.732

Class	Acc.
Angry	15.69%
Disgust	**61.40%**
Fear	**44.23%**
Happy	**39.44%**
Sad	**50.88%**
Neutral	0%
Surprise	0%

Tables 6-7: Model trained on FER2013 and tested with RAF-CE

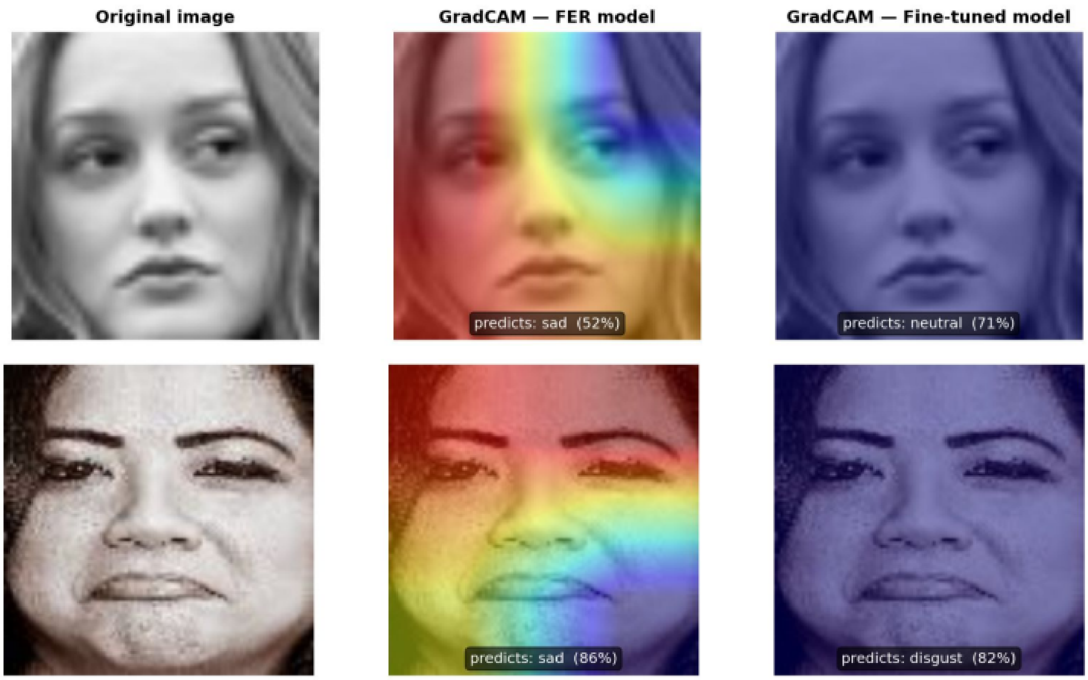
Grad-CAM comparison:

Ground-Truth: Disgust

Original *Grad-CAM before fine-tune* *Grad-CAM after fine-tune*

Figure 8: Improved predictions

Ground-Truth: Sad



Original *Grad-CAM before fine-tune* *Grad-CAM after fine-tune*

Figure 9: Incorrect predictions

Dataset	Scenario	Macro Accuracy	Loss
CK+	Baseline training	99.0%	N/A
FER2013	Full training (30 epochs)	72.47%	0.92
RAF-CE	Baseline (no fine-tuning)	14.32%	6.792
RAF-CE	**After fine-tuning**	30.31%	**1.732**
FER2013	**After fine-tuning**	67.84%	1.9814

Table 8: Summary of all across all datasets

8. Discussion

8.1 Accuracy

For the retraining, on both models, the one trained on CK+ and the one trained on FER2013, we got very similar results to WuJie's original models, where we can clearly see that in comparison, CK+ datasets is much easier to learn compared to FER2013 due to its nature, where CK+ is lab controlled, and FER2013 consists of internet collected faces.

8.2 Grad-CAM

Grad-CAM shows us that the model focuses on different areas depending on the person, in this case, Figure 4, it uses the mouth area to classify the expression, which is a pretty good place for emotion recognition. For figures 6 and 7, both incorrect, we can see two interesting cases. For figure 6, we can see an ambiguous face, where for some people that expression could be happy, while for others it could be neutral. In this case, the model focuses on the eye, which is not the best for expression recognition. In Figure 7, we see an angled face, which was the cause for another incorrect classification, as the model was not sure where to look for classification.

In Figure 5 we can see an example of over-focus on the nose, which is the reason why we decided to do the occlusion experiment.

In Tables 4 and 5 we can see the performance of the model with the occlusion. Overall we see differences in each emotion. For happy, angry and fear, we see a decrease of 10% in the accuracy, which makes sense as we are eliminating part of the face. In disgust and sad emotions, we see a very big difference, which means that the model relies a lot on the nose to classify these emotions. Finally we have neutral and surprise, whose accuracy increased, meaning that it's better for the model to check on other areas, and not so much on the nose.

8.3 Fine-Tune

Before fine-tuning we can see that our model, Table 6, has a very bad performance, an accuracy of 14,32%, which is almost random. After fine-tuning we can see that the performance got better, reaching a 30,31% of accuracy. However, we also see a decrease in accuracy while testing with the FER2013 dataset, Table 8, going from 72,47% to 67,84%.

Some examples were provided, where we can see some cases where the fine-tuning was beneficial, Figure 8, and some cases where it proved detrimental, Figure 9, where the model was correct before the fine-tuning, but classified it incorrectly after it.

Overall the fine-tuning was a success, as we have a more robust model against distinct datasets, even if it means to slightly lower the accuracy on the original dataset.

9. Conclusions

From this project we learned the following things:

- It is important for the dataset to not be perfect, it should contain balanced images, with faces that have angles, noise in the picture, different ethnicities or some more.
- Different emotions rely on different sections of the face for classification.
- Fine-tuning proved effective at making the model more robust against different datasets.
- It is important to have more than one dataset to prepare the model.

Overall, the proposed approach achieved satisfactory results and provided valuable insights into the behavior of facial expression recognition models. However, emotion recognition remains a challenging task because facial expressions can be ambiguous and their interpretation may vary between individuals. Furthermore, emotions are not always experienced independently. In real-life situations, people often exhibit compound or mixed emotions, such as being both surprised and happy simultaneously, whereas the current model is limited to predicting a single emotion category.

Future work should therefore explore the recognition of compound emotions and multi-label classification approaches, which may provide a more realistic representation of human emotional states and improve the applicability of facial expression recognition systems in real-world environments.

10. References

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